Inorganic and Materials Cumulative Exam
April 28 2017

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

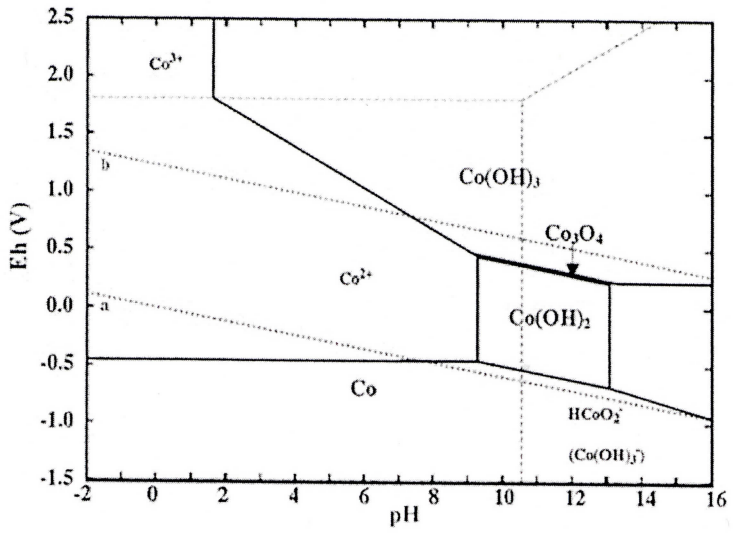

Fig. 1. Pourbaix diagram for Cobalt–H_2O system.

1. Questions regarding the Co-H_2O Pourbaix diagram (Figure 1)
   a. (5pt) What phase(s) of Co is stable in the water window at pH=0?
   b. (5pt) What phase(s) of Co is stable in the water window at pH=9?
   c. (10pt) Experimentally, Co_3O_4 has a band gap of ~1.6eV. Based on information contained in Fig. 1, would it make a good photoelectrode for photoelectrochemical energy conversion (ie., a photocathode for hydrogen evolution or a photoanode for water oxidation)? Please explain your reasoning based on information from the Pourbaix diagram.

2. For semiconductor devices, what is the difference between a heterojunction and a homojunction?

3. What is the role of Cr in the Fe-Cr-C alloy known as ferritic stainless steel? Describe the materials chemistry by which Cr perform that role.

4. Silica gel is a common desiccant used to reduce moisture in containers. How does this work? Please describe the mechanism for desiccation by considering the material properties and morphology in detail.

5. Experimental hypothetical:
    a. (10 pts) Describe two experiments to characterize the surface structure of a gold nanoparticle-poly (ethylene glycol) composite thin film. (An example experiment would look to identify the distribution of gold nanoparticles on the surface of the film.)
    b. (10 pts) Describe two experiments to characterize the bulk structure of a gold nanoparticle-poly (ethylene glycol) composite thin film. (An example experiment would look to identify the distribution of gold nanoparticles on the in the bulk of the film.)

6. There are many ways to classify defects in a crystal lattice. Describe the arrangement of atoms (or lack there of) for the following types of defects:
    a. (5 pts) Substitutional defect
    b. (5 pts) Grain boundary
    c. (5 pts) Interstitial defect
    d. (5 pts) Screw dislocation

7. Decamethylferrocene/Decamethylferrocinium ($Me_{10}Fc^{+/0}$) has been shown to be a better internal standard couple for redox potential in a general solvent than the IUPAC standard ferrocene/ferrocinium ($Fc^{+/0}$). What chemical differences between these two redox couple systems would suggest this to be true?

8. Consider a monatomic crystal formed from a face-centered cubic lattice with a lattice constant of **a**=0.3nm.
    a. (5 pts) What is the distance between an atom and its nearest neighbor?
    b. (5 pts) What is the distance between an atom and its second-nearest neighbor?
    c. (5 pts) If the molar mass of the atomic species is 112g/mol, what is the density of the crystal?
    d. (5 pts) What is d(110), or the spacing between Bragg planes for the hkl=110 Miller index?

9. (12pts) Describe the synthesis of gold nanoparticles from $HAuCl_4$ solutions (sometimes referred to as Turkevich-Frens synthesis). (8pts) How does the addition of a surfactant affect the product of the reaction (ex: CTAB, which specifically binds to a particular gold facet)?

10. (15pts) Describe a Grignard reaction. Draw a schematic organic reaction involving a Grignard reagent. (5 pts) Name two non-carbon elements that are potential targets for reactions with Grignard reagents.